

### 5.1 Quiz: Maclaurin Series — Markscheme

1. [Maximum mark: 7]

AA HL 2025 Nov P1 TZ1 Q05

The first four terms of the Maclaurin series of  $(1 - x)^{-4}$  are  $1 + ax + bx^2 + 20x^3$ .

(a)(i) Show that  $a = 4$ . [2]

**Solution.** Let  $f(x) = (1 - x)^{-4}$ . Then

$$f'(x) = 4(1 - x)^{-5} \mathbf{A1}$$

The coefficient of  $x$  in the Maclaurin series is  $f'(0) = 4(1)^{-5} = 4$ , so  $a = 4$ . **M1 AG**

(a)(ii) Find the value of  $b$ . [3]

**Solution.**

$$f''(x) = 20(1 - x)^{-6} \mathbf{A1}$$

Coefficient of  $x^2$  is  $\frac{f''(0)}{2!} = \frac{20}{2} = 10$ , so

$$b = 10. \mathbf{A1 A1}$$

(Alternative: binomial expansion with negative integer power gives  $\binom{-4}{2}(-x)^2 = \frac{(-4)(-5)}{2}x^2 = 10x^2$ .)

(b) Estimate the value of the car four years ago. [2]

**Solution.** Each year the value is multiplied by 0.9, so the value four years ago,  $V_0$ , satisfies  $V_0(0.9)^4 = 1000$ , i.e.  $V_0 = 1000(0.9)^{-4} = 1000(1 - 0.1)^{-4}$ .

Using the expansion from part (a) with  $x = 0.1$ :

$$(1 - 0.1)^{-4} \approx 1 + 4(0.1) + 10(0.1)^2 + 20(0.1)^3 = 1 + 0.4 + 0.1 + 0.02 = 1.52. \mathbf{M1}$$

$$V_0 \approx \$1520. \mathbf{A1}$$

2. [Maximum mark: 7]

AA HL 2024 May P1 TZ1 Q08

(a)(i) First two non-zero terms of  $\sin(x^2)$ . [2]

**Solution.** Start from  $\sin u = u - \frac{u^3}{3!} + \dots$  and substitute  $u = x^2$ :

$$\sin(x^2) = x^2 - \frac{x^6}{6} + \dots \mathbf{A1 A1}$$

(a)(ii) First two non-zero terms of  $\sin^2(x^2)$ . [3]

**Method 1 (square the series).**

**M1**

$$(\sin x^2)^2 = \left(x^2 - \frac{x^6}{6} + \dots\right)^2 = x^4 - 2 \cdot x^2 \cdot \frac{x^6}{6} + \dots = x^4 - \frac{x^8}{3} + \dots \mathbf{A1 A1}$$

**Method 2 (double-angle identity).** Use  $\sin^2 u = \frac{1 - \cos 2u}{2}$  with  $u = x^2$ :

$$\sin^2(x^2) = \frac{1}{2}(1 - \cos 2x^2) = \frac{1}{2} \left[ \frac{(2x^2)^2}{2!} - \frac{(2x^2)^4}{4!} + \dots \right] = x^4 - \frac{x^8}{3} + \dots$$

**(b)** First two non-zero terms of  $4x \sin(x^2) \cos(x^2)$ . [2]

**Method 1 (derivative recognition).** Note that  $4x \sin(x^2) \cos(x^2) = \frac{d}{dx} [\sin^2(x^2)]$ .  
**M1**

Differentiating the series from (a)(ii):

$$\frac{d}{dx} \left( x^4 - \frac{x^8}{3} + \dots \right) = 4x^3 - \frac{8x^7}{3} + \dots \mathbf{A1}$$

**Method 2 (double-angle).** Use  $4x \sin(x^2) \cos(x^2) = 2x \sin(2x^2)$ , then

$$2x \left[ 2x^2 - \frac{(2x^2)^3}{6} + \dots \right] = 4x^3 - \frac{8x^7}{3} + \dots$$

3. [Maximum mark: 14]

AA HL 2021 Nov P1 TZ0 Q11

**(a)** Prove by induction that  $\frac{d^n}{dx^n}(x^2 e^x) = [x^2 + 2nx + n(n-1)]e^x$  for  $n \in \mathbb{Z}^+$ . [7]

**Solution.**

*Base case*  $n = 1$ :

$$\text{LHS} = \frac{d}{dx}(x^2 e^x) = 2x e^x + x^2 e^x = (x^2 + 2x)e^x. \mathbf{A1}$$

$$\text{RHS} = [x^2 + 2(1)x + 1(0)]e^x = (x^2 + 2x)e^x. \mathbf{A1}$$

So the statement holds for  $n = 1$ .

*Inductive step.* Assume the statement holds for  $n = k$ , i.e.

$$\frac{d^k}{dx^k}(x^2 e^x) = [x^2 + 2kx + k(k-1)]e^x. \mathbf{M1}$$

Differentiating both sides:

**M1**

$$\begin{aligned}\frac{d^{k+1}}{dx^{k+1}}(x^2 e^x) &= \frac{d}{dx} \{ [x^2 + 2kx + k(k-1)]e^x \} \\ &= (2x + 2k)e^x + [x^2 + 2kx + k(k-1)]e^x \mathbf{A1} \\ &= [x^2 + (2k+2)x + (k^2 + k)]e^x \\ &= [x^2 + 2(k+1)x + (k+1)k]e^x \mathbf{A1}\end{aligned}$$

This is the statement with  $n = k+1$ . Since the statement holds for  $n = 1$  and (statement at  $n = k$ )  $\Rightarrow$  (statement at  $n = k+1$ ), by mathematical induction it holds for all  $n \in \mathbb{Z}^+$ . **R1**

**(b)** Maclaurin series of  $f(x) = x^2 e^x$  up to  $x^4$ .

[3]

**Method 1.** Using the formula from part (a) at  $x = 0$ :  $\frac{d^n}{dx^n}(x^2 e^x) \Big|_{x=0} = n(n-1)$ , so

$$f(0) = 0, \quad f'(0) = 0, \quad f''(0) = 2, \quad f'''(0) = 6, \quad f^{(4)}(0) = 12. \mathbf{M1}$$

Therefore

$$f(x) = 0 + 0 + \frac{2}{2!}x^2 + \frac{6}{3!}x^3 + \frac{12}{4!}x^4 + \dots = x^2 + x^3 + \frac{1}{2}x^4 + \dots \mathbf{M1 \ A1}$$

**Method 2.** Multiply the known series for  $e^x$  by  $x^2$ :

$$x^2 \left( 1 + x + \frac{x^2}{2!} + \dots \right) = x^2 + x^3 + \frac{1}{2}x^4 + \dots$$

**(c)** Evaluate  $\lim_{x \rightarrow 0} \frac{(x^2 e^x - x^2)^3}{x^9}$ .

[4]

**Solution.** From (b),  $x^2 e^x - x^2 \approx x^3 + \frac{1}{2}x^4 + \dots$ .

**M1**

So

$$(x^2 e^x - x^2)^3 \approx \left( x^3 + \frac{1}{2}x^4 + \dots \right)^3 = x^9 \left( 1 + \frac{1}{2}x + \dots \right)^3, \mathbf{A1}$$

$$\frac{(x^2 e^x - x^2)^3}{x^9} \approx \left( 1 + \frac{1}{2}x + \dots \right)^3 \rightarrow 1 \quad \text{as } x \rightarrow 0. \mathbf{A1 \ A1}$$

$$\boxed{\lim_{x \rightarrow 0} \frac{(x^2 e^x - x^2)^3}{x^9} = 1.}$$